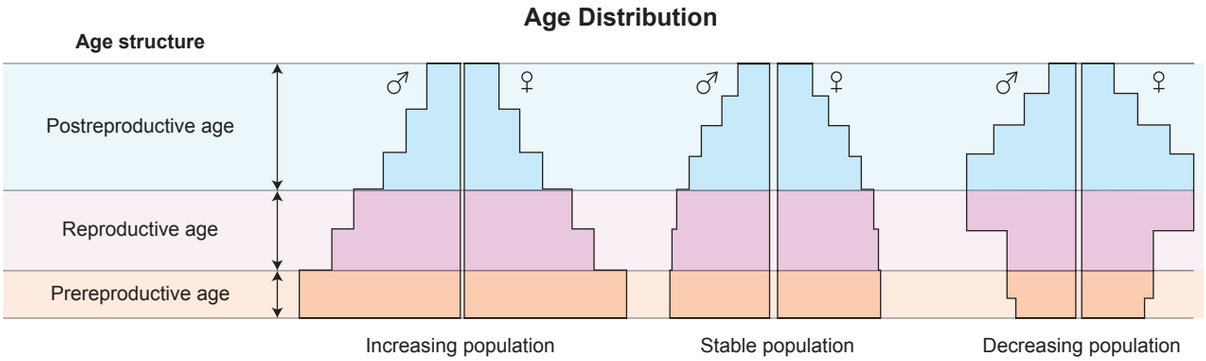


Fig. 10.10 Age distribution pyramids

over exploitation of the habitat. Further, it leads to occupation of new areas elsewhere.

Immigration

It leads to a rise in population levels. If the population increases beyond the carrying capacity, it can result in increased mortality among the immigrants or decreased reproductive capacity of the individuals.

Both emigration and immigration are initiated or triggered by weather and other abiotic and biotic factors.

10.9 Population Age Distribution:

The proportion of the age groups (pre-reproductive, reproductive and post reproductive) in a population is its age distribution attribute. This determines the reproductive status of the population at the given time and is an indicator of the future population size.

Usually a rapidly growing population will have larger proportion of young individuals. A stable population will have an even distribution of various age classes. A declining population tends to have a larger proportion of older individuals (**Fig. 10.10).**

10.10 Growth Models / Curves

Populations show characteristic growth patterns or forms. These patterns can be plotted and termed as J-shaped growth form and S-shaped growth form (Sigmoid form).

J-Shaped growth form

When a population increases rapidly in an exponential fashion and then stops abruptly due to environmental resistance or due to sudden appearance of a limiting factor, they are said to exhibit J-shaped growth form. Many insects show explosive increase in number during the rainy season followed by their disappearance at the end of the season (**Fig. 10.11).**

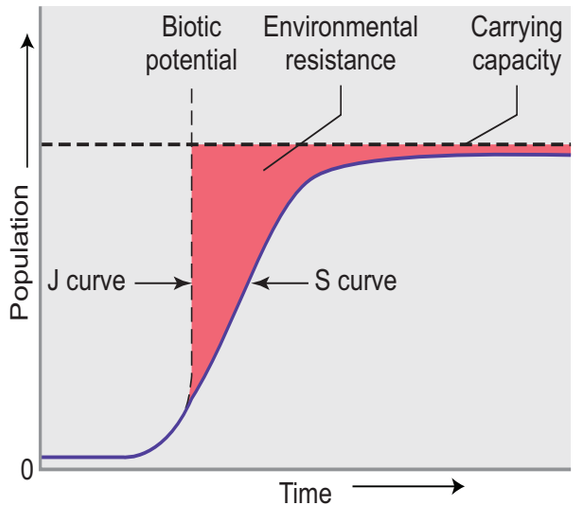


Fig. 10.11 J-Shaped and S-Shaped growth curves

S-Shaped growth form (sigmoid form)

Some populations, as in a population of small mammals, increase slowly at first then more rapidly and gradually slow down as environmental resistance increases whereby equilibrium is reached and maintained. Their

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